

African
Brain
Data
Network

Module 1 Introduction to Electrophysiology

Recording techniques used in Neuroscience

30th of July 2026 – Nicole Vissers



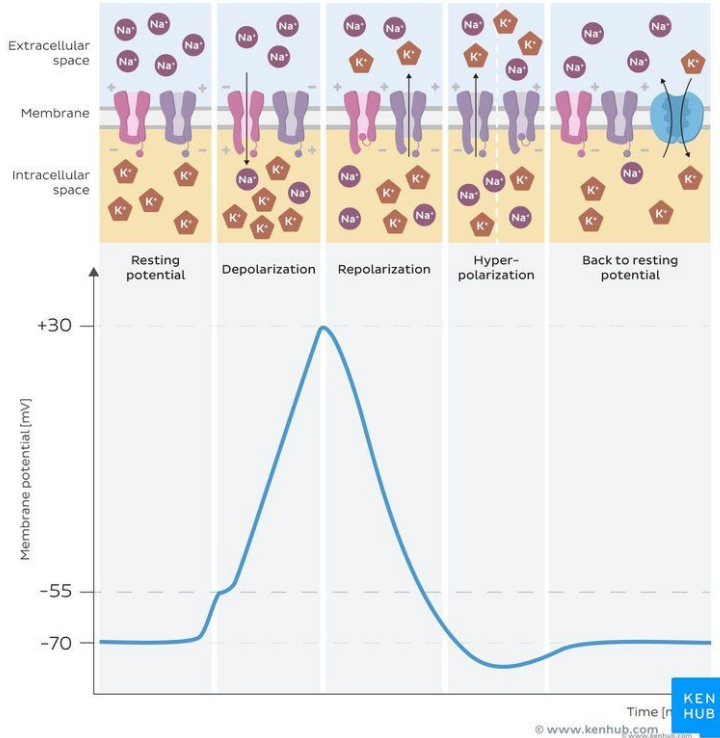
Discuss Assignment Last Week!

Write a paragraph on:

What is a new fact you have learned about the brain?

What questions arose when learning about the brain?

Recap from last week!



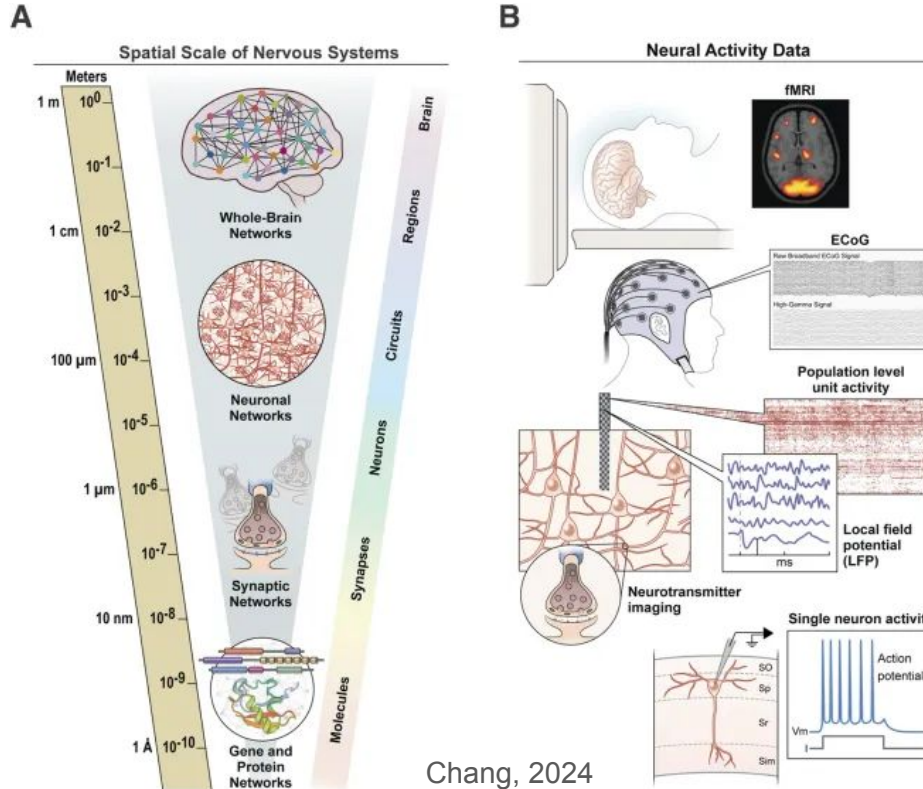
1. The brain consists of highly connected **neurons**
2. These neurons have a membrane potential that changes by **ion flow**
3. A change in ion flow results in change in **voltage** that can be measured
4. When the membrane potential reaches a threshold it fires an **action potential**
5. An action potential results in **neurotransmitter release**
6. Neurotransmitter release affects the membrane potential of other neurons
7. The **type** of neurotransmitter affects the eventual result

We can study this process at different scales and using various techniques.



"If neurons communicate using electrical signals, how can we listen to them?"

Every scale requires different techniques!



From small to bigger scale:

Intracellular recordings

- Whole-cell recording
- Cell-attached recording

Extracellular recordings

- Single-unit recordings
- Local field potentials
- Neuropixels recordings

Whole-brain recordings (basically also extracellular)

- EEG
- MEG

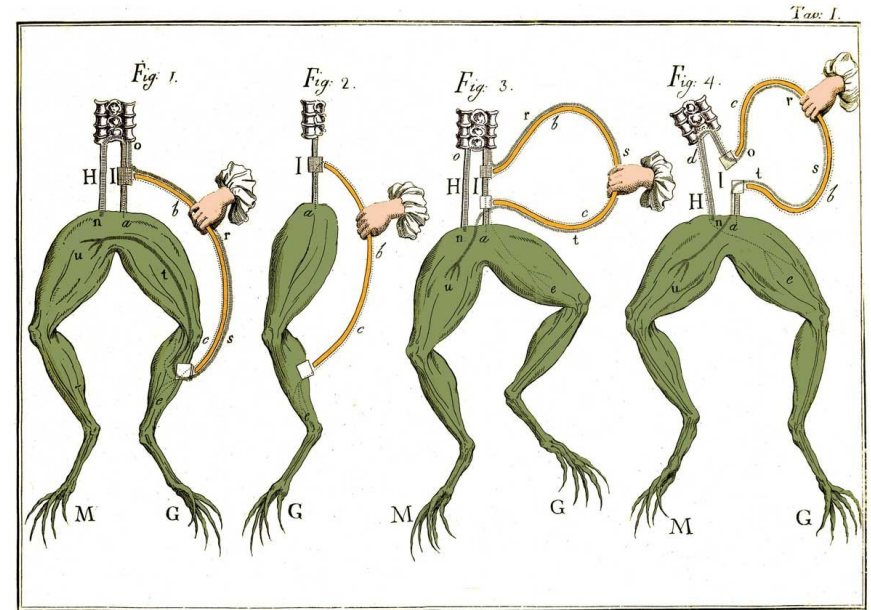
Not using electrophysiological phenomena; (f)MRI & fNIRS

Electrophysiology in the 18th century

Luigi Galvani in 1791

Frog muscle contracts when the leg nerve and muscle were connected through a metal conductor;

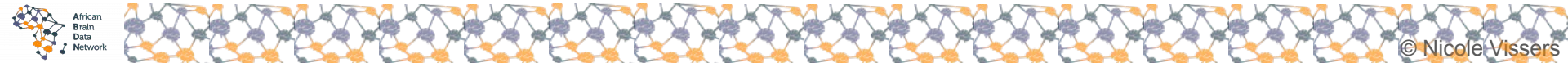
Animal Electricity





Intracellular recordings

What is happening inside one neuron?



One of the first intracellular recordings

Hodgkin & Huxley (1939)

Record electrical pulses in the nerve fibres of the Squid! They proved that electrical impulses overshoot the resting potential.

Their **mathematical formulation** of the nerve impulses (1952) gave them **Nobel Prize** in Physiology or Medicine (1963)

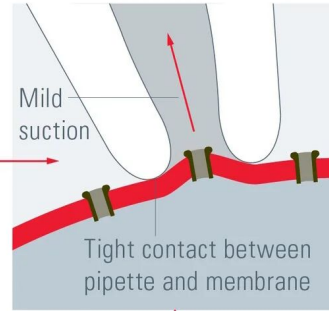
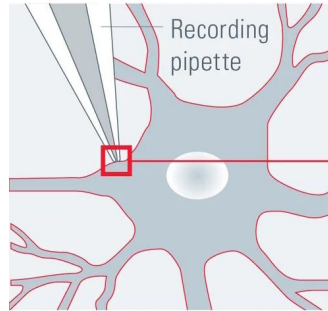


Intracellular recordings

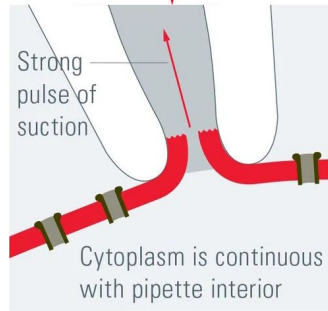
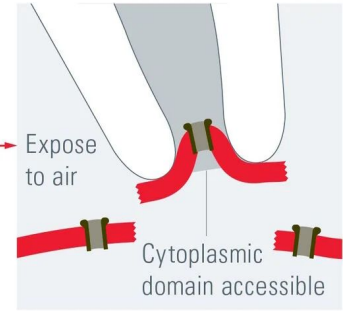
Record from within a neuron;

- Cell-attached recording
 - Inside-out recording
 - Whole-cell recording
 - Outside-out recording

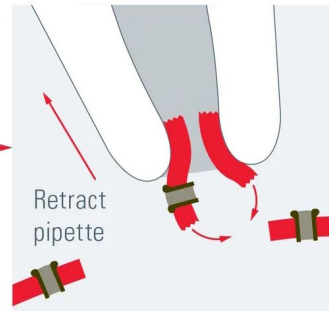
Cell-attached recording



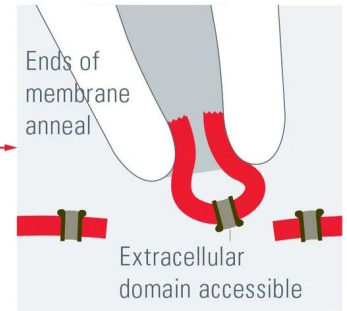
Inside-out recording



Whole-cell recording



Outside-out recording



Leica Microscopes



What does it measure?

Measures electrical properties of individual neurons:

- Membrane potential
- Synaptic inputs
- Ionic current
- Action Potentials



Pros and Cons;

Advantaged

- Direct measurements of:
 - EPSPs & IPSPs
 - Action potentials
- Measures subthreshold events (millivolts!)
- Can manipulate the neuron
- High temporal precision (microsecond!)
- Study cellular mechanisms:
 - Ion channels
 - Neurotransmitter receptors
 - Synaptic transmission
 - Intrinsic excitability

Disadvantages

- Difficult
- One neuron at a time
- Invasive to the neuron; can die over time so can't record long
- Difficult in freely moving animals



Try these tools yourself:

Virtual Patch Clamp Simulator (University of Michigan):

<https://compbio.fhs.um.edu.mo/virtualpatch/>

Model the Hodgkin-Huxley neuron:

<https://docs.neuroml.org/Userdocs/SingleCompartmentHHExample.html>



Extracellular recordings

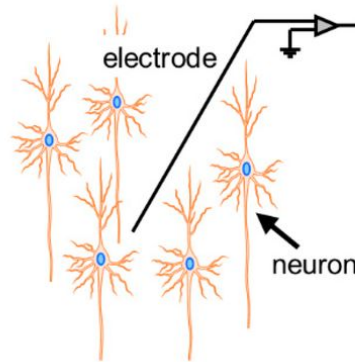
What are neurons firing in a circuit?

What does it measure?

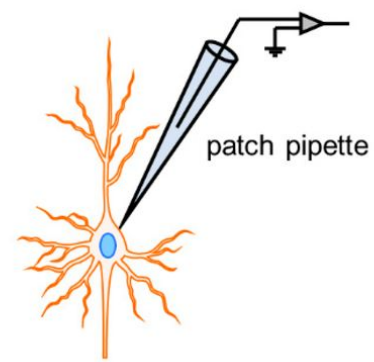
Electrical activity outside of the neurons

- Action potentials
- Population activity
- Local field potentials

(a) Extracellular recording

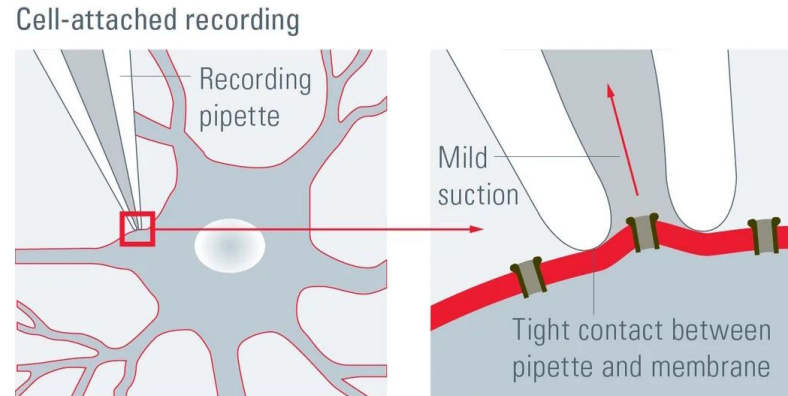


Intracellular recording



Single-unit recordings

Voltage changes outside the cell; so the membrane doesn't break



Leica Microscopes

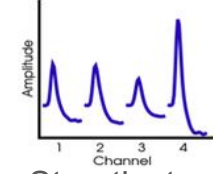
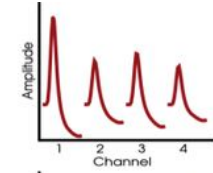
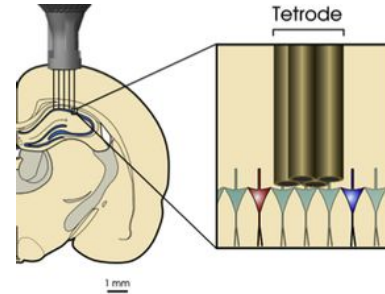
Multi-unit recordings

Tetrodes

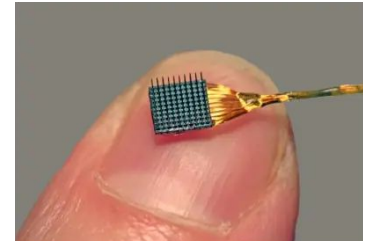
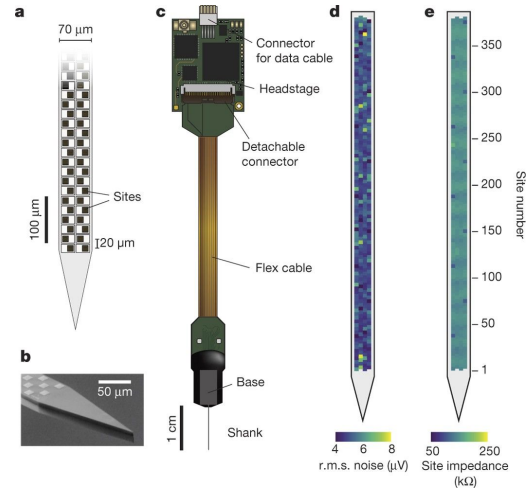
Silicon probes

Neuropixels

Multi-electrode arrays



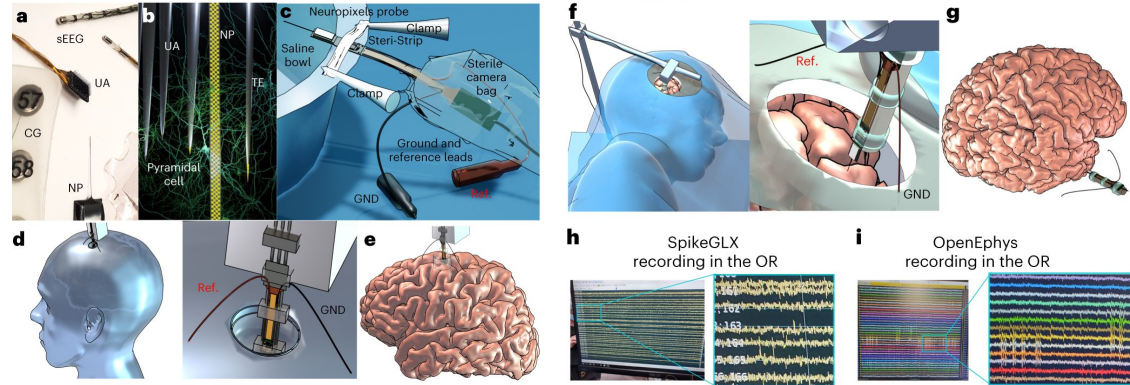
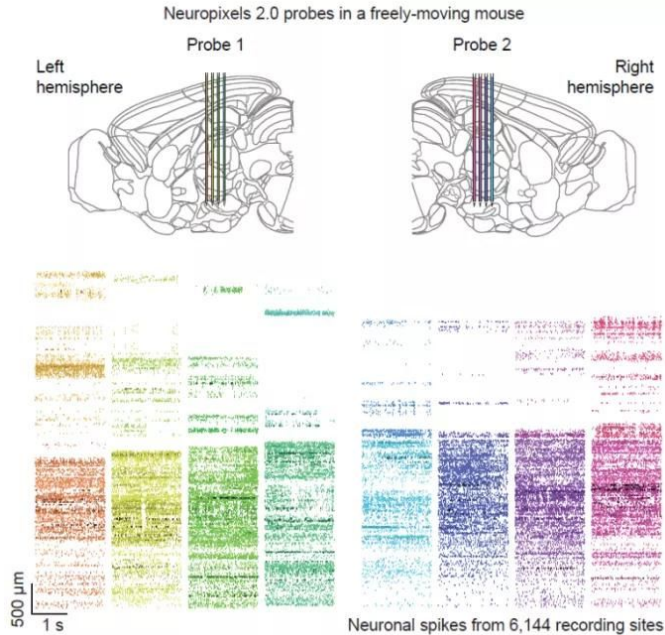
Stuartlayton



Brain Latam

Jun et al., 2017

Neuropixels



Coughlin et al., 2023

Free online course to cover everything:

<https://www.ucl.ac.uk/brain-sciences/research/neuropixels/training/2025-neuropixels-course>



Pros and Cons; Extracellular recordings

Advantages

- Many neurons at the same time
- Can be done in freely-moving settings
- Less invasive than intracellular
- Can record relatively long
- Great temporal resolution
- Record across multiple brain regions

Disadvantages

- Does not measure membrane potential; only spikes
- More susceptible to noise
- Cell-identify is uncertain
- More complicated analyses

Ideal to study underlying brain circuits

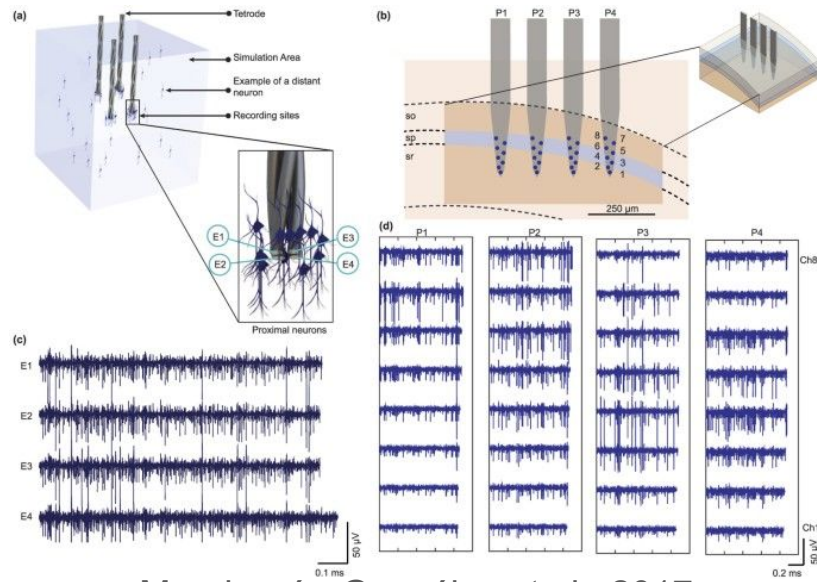


Whole-brain recordings

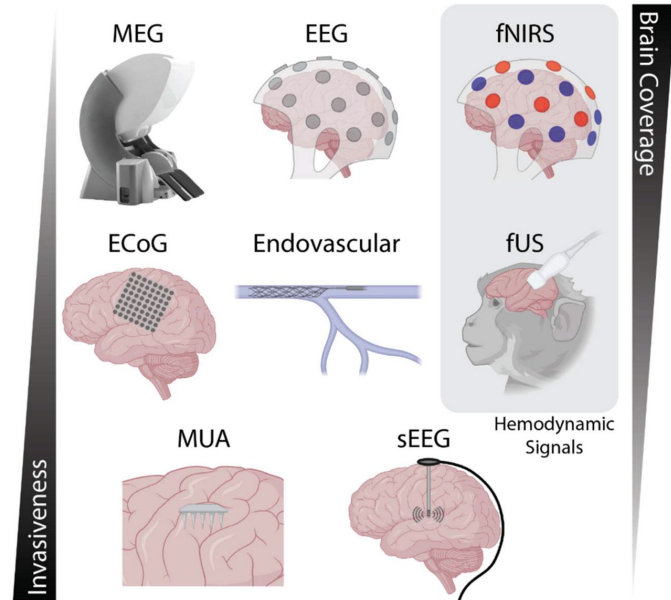
What is the activity of large populations across the brain?

What does it measure?

Synchronous activity from large populations of neurons

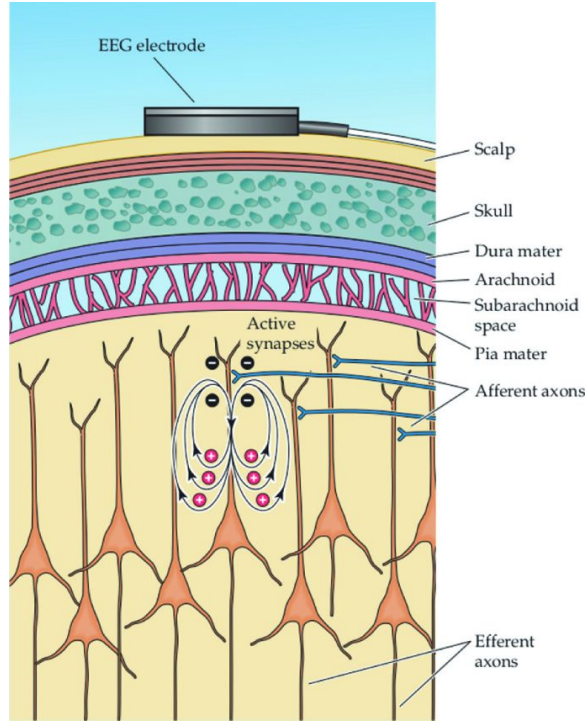


Mondragón-González et al., 2017



Edelman et al., 2025

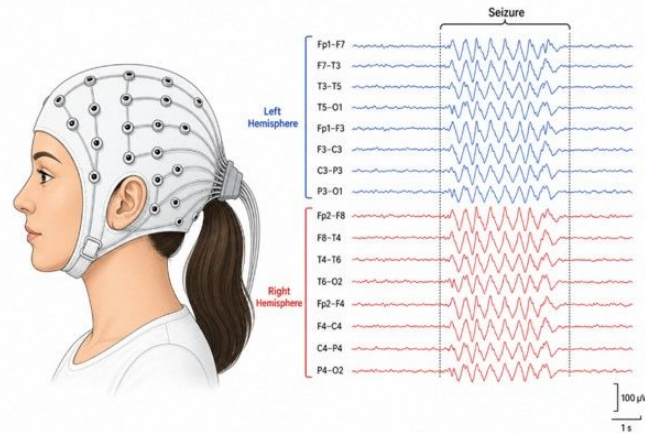
Electroencephalogram (EEG)



Purves

Non-invasive brain imaging method

Measures electrical currents in the brain



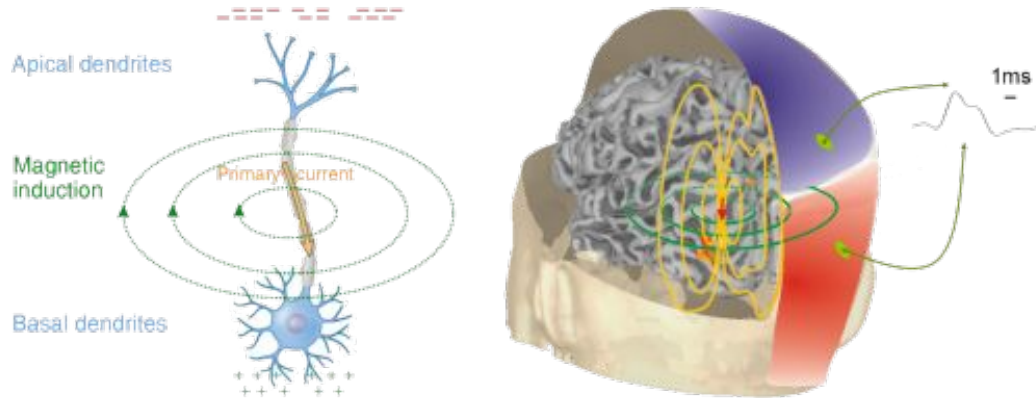
An electroencephalogram (EEG)

BIOserenity

Magnetoencephalography (MEG)

Non-invasive brain imaging method

Measures magnetic fields that occur by the electrical currents in the brain



McGill



NIMH



Pros and Cons; Whole-brain recordings

Advantages

- Non-invasive & relatively safe
- Can be used in humans
- Great temporal resolution
- Relatively inexpensive
- Large spatial coverage

Disadvantages

- Poor spatial resolution
- Measures summed-up activity
- Quite sensitive to artefacts (motion, sweat)
- Hard to find the actual source
- Hard to record from deeper brain structures
- MEG is very expensive

Can have a clinical benefit for epilepsy, sleep, coma and brain-computer interfaces.



What to prepare for next week!

Use github:

https://github.com/African-Brain-Data-Network/ABDN_2026_FELLOWSHIP_COHORT/tree/main/participants

Write a paragraph on:

What is your preferred recording technique? And why?

If you could design any experiment, what would you do?